



When Are Text/LLM Embeddings Valid for Causal Adjustment?

Embedding-Based Causal Adjustment in PSID
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Why Embedding-Based Adjustment?

Estimating the causal effect of job-related training on household earnings from observational panel data is fundamentally difficult. People who choose training differ from those who do not in many ways at once: some are recorded in survey data (e.g., income history, education), but others—motivation, labor-market attachment, social networks—are latent and rarely measured. Without adjusting for these latent traits, an observed earnings gap mixes the genuine effect of training with pre-existing differences between participants and non-participants.

Causal Identification on PSID Panel Data

Sample: 6,107 households (1,222 trained, ~20%) with valid pre-/post-treatment data across waves.

Pre-treatment window: Covariates drawn from PSID waves 2007, 2009, and 2011 (strictly before treatment).

Treatment (T_i): Participation in job-related training during the 2013 and/or 2015 waves.

Outcome (Y_i): Post-treatment log(total family income) from the 2019 and 2021 waves

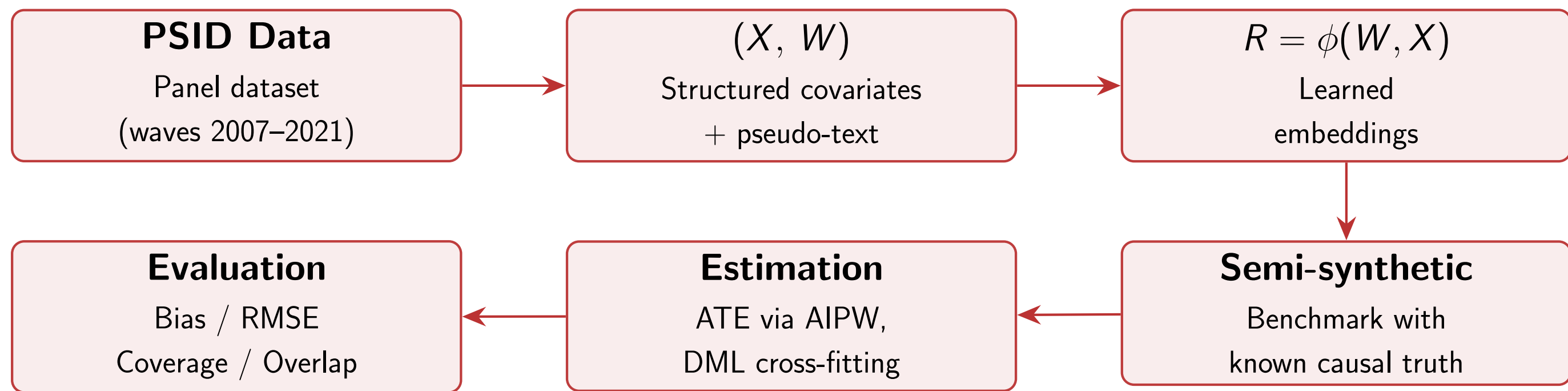
Target estimand: Average Treatment Effect at horizon h ,

$$ATE_h = \mathbb{E} \left[Y_i^1(t+h) - Y_i^0(t+h) \right].$$

Identification: Conditional ignorability ($Y^0, Y^1 \perp T \mid Z$ and positivity $0 < P(T=1 \mid Z) < 1$).

X	Examples
Demographics	Age, sex, race, education of household head
Household composition	Family size, children, marital status
Economic status	Labor income, total family income, financial support
Housing and assets	Home value, mortgage, savings, stocks, retirement accounts
Employment history	Employment status, years worked, business ownership
Health and behavior	Chronic conditions, distress, smoking, alcohol use

Overall Workflow



Pseudo-text Construction

Each household first receives a deterministic fact list from strictly pre-treatment covariates:

$$F_i = g(X_i^{pre}).$$

We then optionally use an LLM as a constrained rewrite function to convert facts into fluent pseudo-text:

$$W_i = \text{LLM}(F_i; \text{facts-only prompt}),$$

where the prompt forbids inference and post-treatment information. Final embedding is the template note W_i .

Example pseudo-note (W_i): *[DEMO] age: 57; hh.size: 1; sex: female; race: white; marital: divorced; education: lt.hs. [EMP] work_for.money.now: no; years.worked.since.18: 25.years; labor.income.2014: none; family.income.2014: mid. [HEALTH] self.rated.health: fair; K6.distress: moderate. [ASSET] house.value: not.homeowner; mortgage: not.homeowner; business.owner.2014: no; WIC.2014: ineligible; checking.saving.2015: none; stocks.2015: none.or.break.even; IRA.2015: none; credit_card.debt.2015: none; census.needs.2014: mid.low.*

Embedding Pipelines

All pipelines map pseudo-text W_i and covariates X_i to a d -dimensional representation, fitted with cross-fitting:

$$R_i = \phi(W_i, X_i) \in \mathbb{R}^d.$$

- **A. TF-IDF + SVD:** Sparse n -gram features $\psi(W_i)$ reduced via truncated SVD; encoder-free baseline.
- **B. Sentence-Transformer + PCA:** Pretrained transformer (e.g., MiniLM) followed by L_2 -normalization and PCA; semantic but single-wave.
- **C. Trajectory Embedding:** Concatenate prior notes before encoding to capture temporal dynamics:

$$R_i = \text{PCA}_k \left(L_2(f_\theta(W_i^{\text{traj}})) \right), \quad W_i^{\text{traj}} = \text{concat}(W_{i,t-K}, \dots, W_{i,t-1}).$$

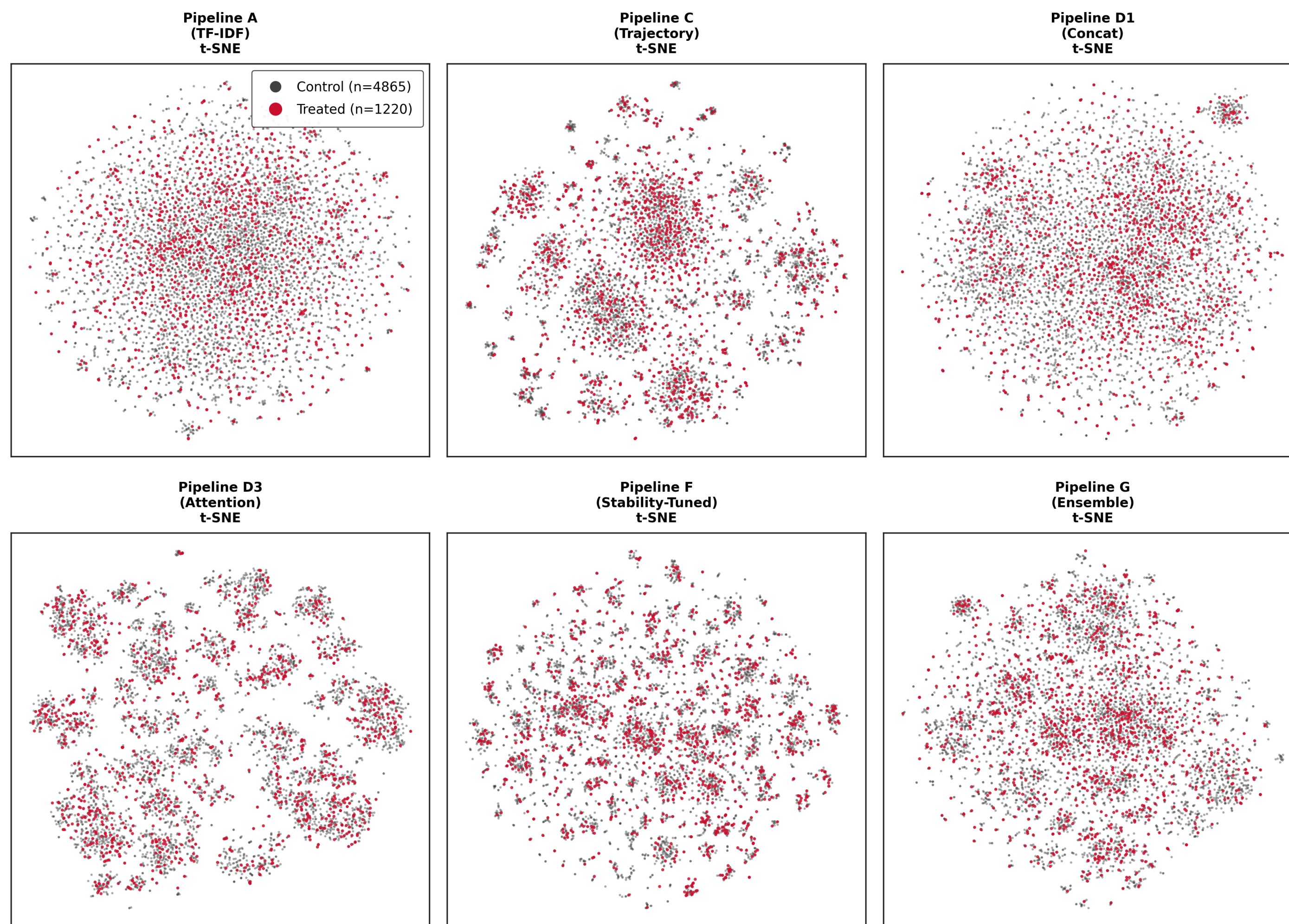
- **D. Multi-view Fusion:** Encode work-history ($E^{(1)}$) and household-context ($E^{(2)}$) separately, then combine: $D1$ concat + PCA (no learned fusion); $D2$ learned linear $A_1 E^{(1)} + A_2 E^{(2)}$ with ridge; $D3$ attention-weighted convex combination via softmax scorer $g(\cdot)$.
- **E. Overlap-aware Projection:** Adversarial min-max objective (gradient reversal) preserving outcome signal while making T unpredictable from PE_i :

$$\min_{P, \beta} \mathcal{L}_Y(P, \beta) - \gamma \mathcal{L}_{\text{ovlp}}(P, \omega).$$

- **F / G. Stability-Tuned / Ensembled:** F applies contrastive InfoNCE over benign W_i augmentations; G stacks heterogeneous encoders $f_{\theta_1}, f_{\theta_2}$. Both reduce idiosyncratic embedding geometry. F's objective:

$$\mathcal{L}_{\text{NCE}} = - \sum_{i \in B} \log \frac{\exp(\langle \hat{u}_i^{(1)}, \hat{u}_i^{(2)} \rangle / \tau)}{\sum_{k \in B} \exp(\langle \hat{u}_i^{(1)}, \hat{u}_k^{(2)} \rangle / \tau)}.$$

Examples of Embedding Pipelines Visualizations



Semi-synthetic Benchmark

Latent Confounder $U_{i,t-1} = \text{std}(w^\top H_{i,t-1}) + \sigma_u \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1),$

Treatment $P(T_i = 1 \mid \tilde{X}_i, U_i) = \sigma \left(\alpha_0 + s_x \alpha_x^\top \tilde{X}_i + s_u \alpha_u U_i \right),$

Potential Outcomes $Y_i^0 = g(\tilde{X}_i) + \gamma_u U_i + \rho Y_i^{\text{base}} + \varepsilon_i, \quad Y_i^1 = Y_i^0 + \tau(\tilde{X}_i, U_i),$

Benchmark difficulty spans *linear* to *threshold* regimes for $g(\cdot)$ and *constant* to *nonlinear-heterogeneous* regimes for $\tau(\cdot)$:

$$g_{\text{linear}}(\tilde{X}) = \beta^\top \tilde{X}, \quad g_{\text{threshold}}(\tilde{X}) = \beta^\top \tilde{X} + c \mathbf{1}\{\tilde{X}_m < 0\} + d \max(0, \tilde{X}_n - 1),$$

$$\tau_{\text{constant}} = \tau_0, \quad \tau_{\text{nonlinear}}(\tilde{X}, U) = \tau_0 + \tau_1 \tanh(\tilde{X}_m) + \tau_2 \mathbf{1}\{\tilde{X}_n > 0\} + \tau_3 U^2.$$

Causal Estimation Methods

For each adjustment regime $Z \in \{X, R, [X; R]\}$, we estimate the treatment effect using:

- Difference in means: raw treated-control outcome gap.
- Regression adjustment: separate outcome models $\hat{\mu}_1(Z)$ and $\hat{\mu}_0(Z)$.
- IPW / Hájek IPW: propensity-score weighting using $\hat{e}(Z)$.
- AIPW: doubly robust combination of outcome and propensity models.
- DML-AIPW: cross-fitted version using out-of-sample nuisance predictions.

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{T_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{e}(Z_i)} \right].$$

Results

Tier	Best R only RMSE	Best pipeline	X only RMSE	Reduction
0	0.44	G-ENSSTK-50	1.12	2.5×
1	0.58	D2-LINF-50	1.95	3.4×
2	0.50	G-ENSSTK-50	1.21	2.4×
3	0.55	G-ENSSTK-50	1.28	2.3×

Table: Across all four tiers and all three outcome scenarios, the '(R)' condition achieves lower RMSE than '(X)' and '(XR)'. The best embedding pipeline reduces RMSE by 2–3× relative to the covariate-only baseline.

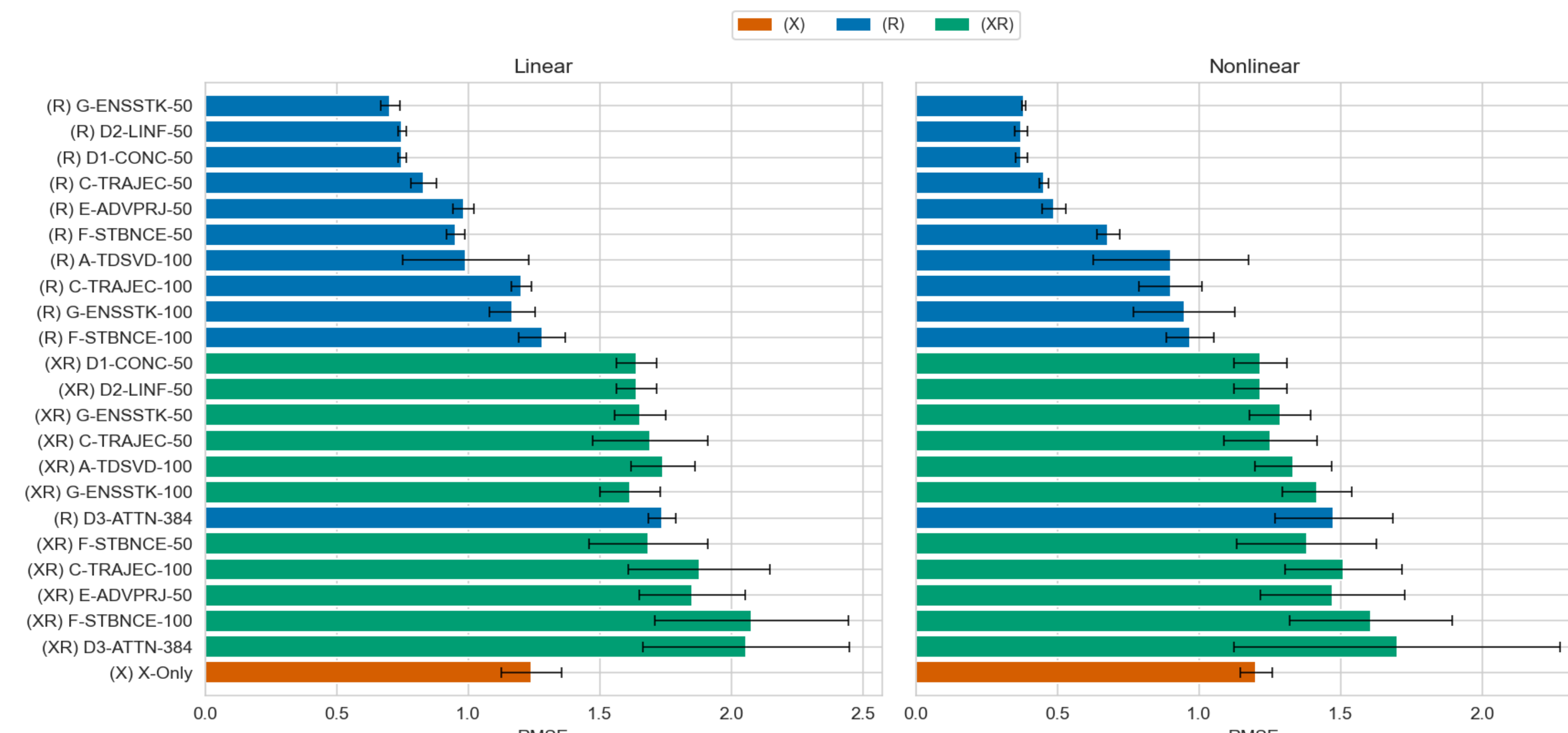


Figure: RMSE comparison across tiers and adjustment settings.

Results

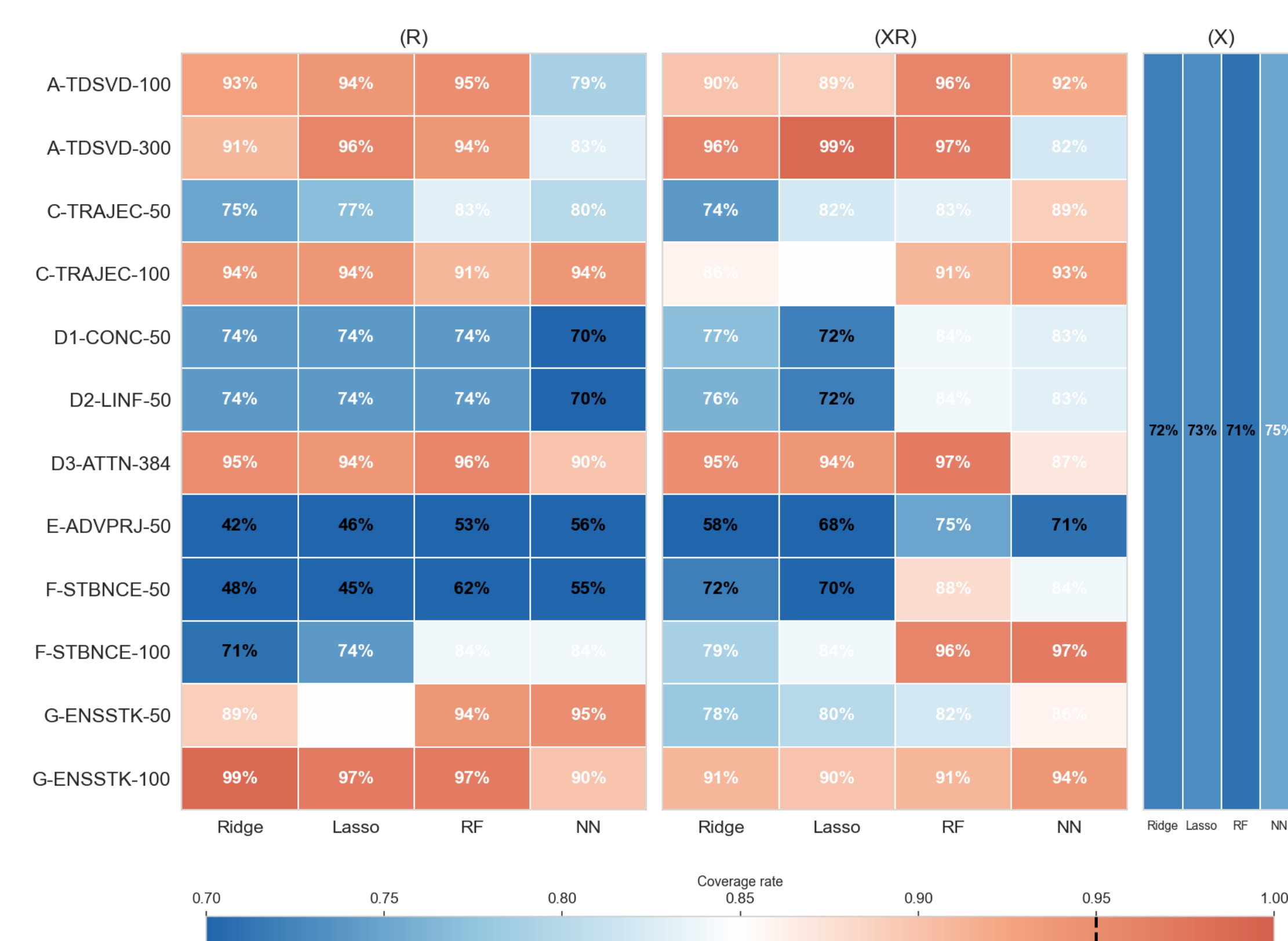


Figure: More Coverage in X+R! but better coverage in R.

(R) A-TDSVD-100	0.655	1.060	0.699	0.726
(R) A-TDSVD-300	2.285	4.116	2.487	2.882
(R) C-TRAJEC-50	0.706	0.626	0.643	0.724
(R) C-TRAJEC-100	0.935	0.903	0.924	0.993
(R) D1-CONC-50	0.655	0.663	0.642	0.653
(R) D2-LINF-50	0.654	0.664	0.641	0.653
(R) D3-ATTN-384	1.383	1.501	1.396	1.355
(R) E-ADVPRJ-50	0.931	0.856	0.868	0.941
(R) F-STBNCE-50	0.847	0.839	0.785	0.855
(R) F-STBNCE-100	1.021	0.962	0.947	1.129
(R) G-ENSSTK-50	0.558	0.584	0.517	0.579
(R) G-ENSSTK-100	0.861	1.039	0.839	0.963
(XR) A-TDSVD-100	1.375	1.489	1.326	1.350
(XR) A-TDSVD-300	1.866	2.843	1.925	2.026
(XR) C-TRAJEC-50	1.435	1.315	1.159	1.413
(XR) C-TRAJEC-100	1.710	1.291	1.395	1.591
(XR) D1-CONC-50	1.330	1.206	1.244	1.333
(XR) D2-LINF-50	1.330	1.209	1.242	1.333
(XR) D3-ATTN-384	1.428	2.211	1.518	1.443
(XR) E-ADVPRJ-50	1.543	1.371	1.258	1.526
(XR) F-STBNCE-50	1.371	1.153	1.088	1.292
(XR) F-STBNCE-100	1.619	1.636	1.217	1.637
(XR) G-ENSSTK-50	1.390	1.235	1.215	1.336
(XR) G-ENSSTK-100	1.352	1.200	1.207	1.349
X-Only	0.869	0.999	1.019	0.829

Figure: R is better!

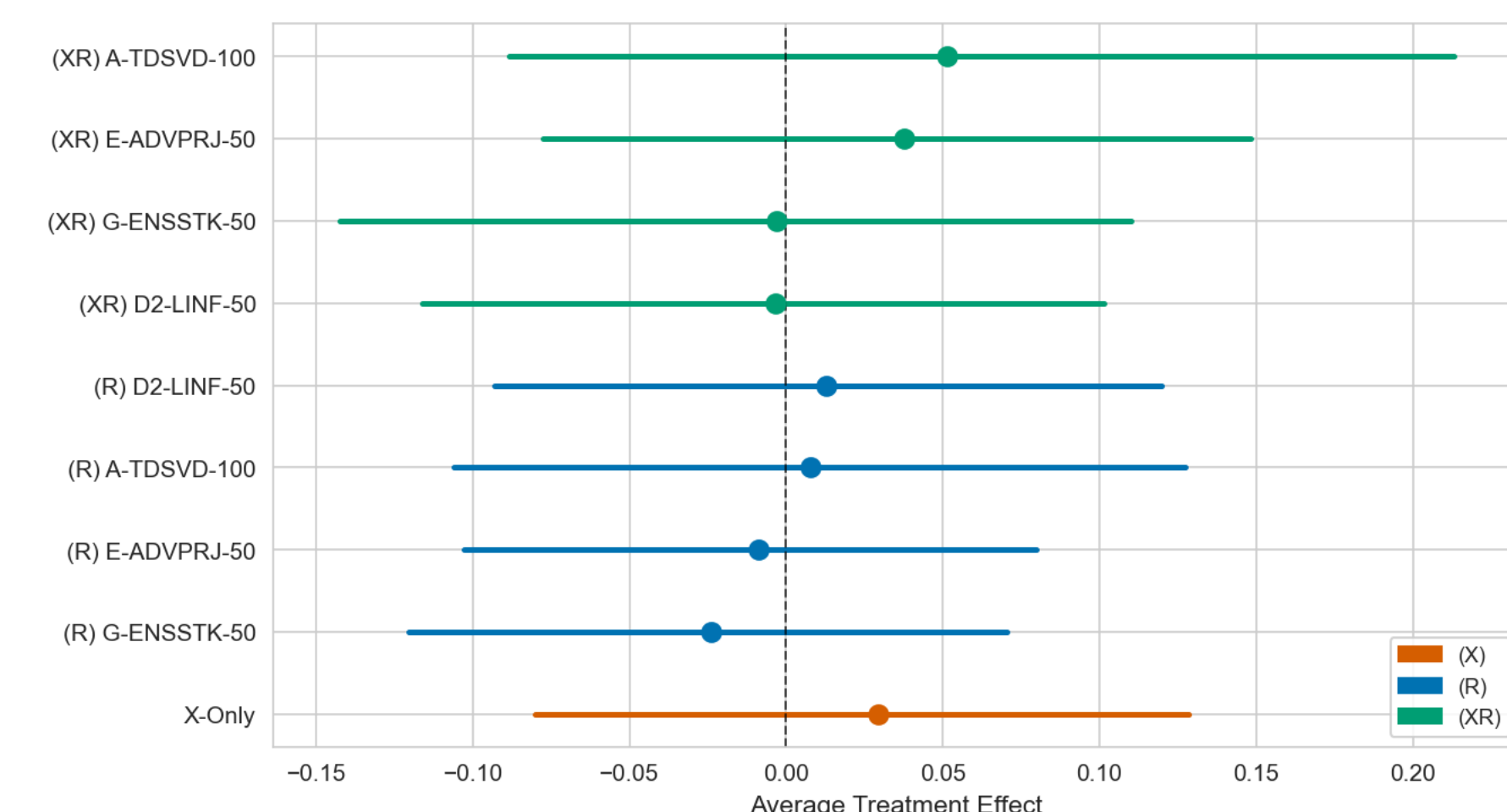


Figure: Job training has no significant impact on real household income

Conclusion and Future Work

- **Embeddings > Covariates:** R reduces RMSE by 2x to 3.4x compared to X . The best R reduced bias by 55% compared to X and XR .
- **The Coverage Tradeoff:** R offer the **best point accuracy**, while XR trade some accuracy for wider confidence interval coverage.
- **Real-World Impact:** In the PSID dataset, the true ATE of job training is near zero; it has no significant impact on total household income.
- **Bayesian uncertainty propagation:** Extend the estimation framework using Bayesian methods to better capture and propagate the representational uncertainty from the embedding generation step into the final causal intervals.

References

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